

An electric fuse is a safety device used to prevent damage to an electrical circuit when excessive current flows through it. The electric fuse works on the principle of the heating effect of electric current. It consists of a piece of wire made of a metal or an alloy of low melting point. As the current increases beyond a limit, the wire in the electric fuse melts and breaks off. The fuse is then said to have blown off. The circuit is broken and current stop flowing through it. Thus, a fuse prevents fires.

17. An electric fuse works on the

- (a) heating effect of current (b) chemical effect of current
(c) magnetic effect of current (d) lighting effect of current

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Ans. (a)

An electric fuse works on the heating effect of electric current. When a current more than the rated amount passes through a fuse, it melts due to excessive heating and breaks the circuit and current stops flowing. It is placed in series with the device and normally it has a low melting point.

18. The wire generally used for installation of safety fuse is made of which material?

- (a) Tin (b) Lead
(c) Nickel (d) An alloy of tin and lead

46th B.P.S.C. (Pre) 2004

Ans. (d)

A fuse is a small piece of wire. It is made of Tin- Lead alloy, its melting point is low. When the current in the circuit exceeds the specified value due to short circuiting, overloading, voltage surge etc., the fuse melts and breaks. Due to this, damage to electric / electronic equipment and any kind of accident is averted.

19. How does the white light produce in the tube?

- (a) by heating up the copper wire
(b) by heating up the filament
(c) by accelerating the atoms
(d) by oscillating the molecules

47th B.P.S.C. (Pre) 2005

Ans. (d)

A tube light or fluorescent lamp is a kind of light that uses a combination of electricity and mercury vapour to create light energy. The electricity causes electrons to travel at high speed between the poles. These electrons hit the mercury gas molecules and oscillate them. The mercury gas is then ionized by the current flowing within, producing ultraviolet radiation. Since the human eye is unable to see this radiation, the glass tube is coated with a fluorescent material which changes it into white light. Thus, among the given options, option (d) is the correct answer.

20. The diode is used as

- (a) an amplifier (b) a rectifier
(c) an oscillator (d) More than one of the above
(e) None of the above

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Ans. (b)

The diode is a semiconductor two terminal electronic component with PN junction. The anode which is the positive terminal of a diode is represented with A and cathode, which is the negative terminal is represented with K. As diode is used as a rectifier, it is also known as rectifier component. We use rectifier to convert an alternating current into a direct current using one or more P-N junction diodes.

Conductivity

1. Due to temperature variation along a conductor, potential variation occurs along it. This phenomenon is known as

- (a) Thomson effect (b) Joule effect
(c) Seebeck effect (d) Peltier effect
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

Due to the temperature difference between the two points of the same conductor, there is a difference in the electron density. The electron density is higher at low temperatures than at higher temperatures. This will be a potential difference between two points of the same conductor. Therefore, when a current (charge) flow in the conductor, work is done against or along the direction of potential difference, due to the flow of current. As a result of this heat is observed or evolved. This effect is called the Thomson effect.

2. The electrical conductivity of a metal decreases with increasing temperature, because

- (a) the energy of electrons increases with increasing temperature
(b) metal increases on heating
(c) metal atoms vibrate more at high temperature
(d) metals have low specific heat
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (e)

The thermal conductivity of any material is dependent on two things :
(i) The motion of free electrons
(ii) Molecular vibrations